

ONLINE MASTERS IN DATA SCIENCE

DSC 257R - UNSUPERVISED LEARNING

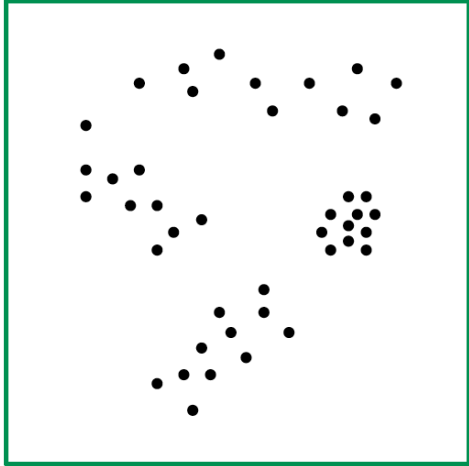
# MIXTURES OF GAUSSIANS

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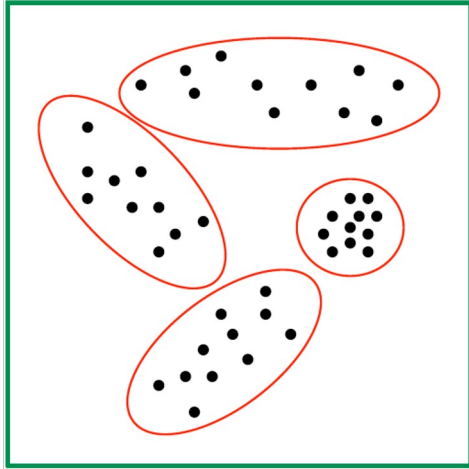
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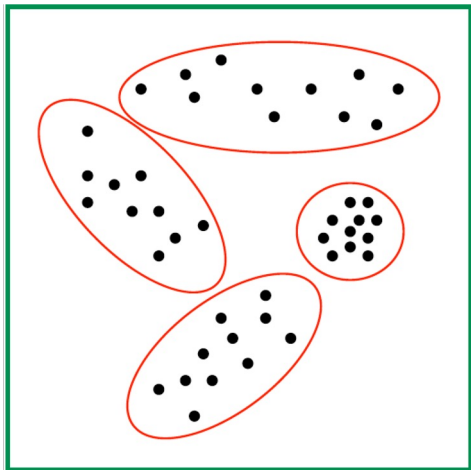
## Mixtures of Gaussians



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Each of the  $k$  clusters is specified by:

- a Gaussian distribution  $P_j = N(\mu_j, \Sigma_j)$
- a mixing weight  $\pi_j$ .

Overall distribution over  $\mathbb{R}^d$ : a **mixture of Gaussians**

$$\Pr(x) = \pi_1 P_1(x) + \cdots + \pi_k P_k(x)$$

## The Generative Model

A mixture of  $k$  Gaussians:

$$\pi_1 N(\mu_1, \Sigma_1) + \pi_2 N(\mu_2, \Sigma_2) + \cdots + \pi_k N(\mu_k, \Sigma_k)$$

The generative process:

- Pick a source  $Z \in \{1, \dots, k\}$  according to distribution  $\{\pi_1, \dots, \pi_k\}$
- Pick  $X \sim N(\mu_z, \Sigma_z)$

This is a **latent variable model**: we observe  $X$  but not  $Z$ .

Given a point  $x$ , what is its most likely label?

## Fitting a Mixture of Gaussians

Given: data  $x_1, \dots, x_n \in \mathbb{R}^d$ .

For any mixture model  $\pi_1, \dots, \pi_k$ ,  $P_1 = N(\mu_1, \Sigma_1), \dots, P_k = N(\mu_k, \Sigma_k)$ ,

$$\begin{aligned} & \Pr(\text{data} \mid \pi_1 P_1 + \dots + \pi_k P_k) \\ &= \prod_{i=1}^n (\pi_1 P_1(x_i) + \dots + \pi_k P_k(x_i)) \\ &= \prod_{i=1}^n \left( \sum_{j=1}^k \frac{\pi_j}{(2\pi)^{d/2} |\Sigma_j|^{1/2}} \exp \left( -\frac{1}{2} (x_i - \mu_j)^T \Sigma_j^{-1} (x_i - \mu_j) \right) \right) \end{aligned}$$

Find the **maximum-likelihood mixture of Gaussians**:

parameters  $\{\pi_j, \mu_j, \Sigma_j : j = 1 \dots k\}$  maximizing this function.

## Optimization Surface

Minimize the negative log-likelihood,

$$\sum_{i=1}^n \ln \left( \sum_{j=1}^k \frac{\pi_j}{(2\pi)^{d/2} |\Sigma_j|^{1/2}} \exp \left( -\frac{1}{2} (x_i - \mu_j)^T \Sigma_j^{-1} (x_i - \mu_j) \right) \right)$$

## The EM (Expectation Maximization) Algorithm

- 1 Initialize  $\pi_1, \dots, \pi_k$  and  $P_1 = N(\mu_1, \Sigma_1), \dots, P_k = N(\mu_k, \Sigma_k)$ .
- 2 Repeat until convergence:

- Assign each point  $x_i$  fractionally between the  $k$  clusters:

$$w_{ij} = \Pr(\text{cluster } j \mid x_i) = \frac{\pi_j P_j(x_i)}{\sum_{\ell} \pi_{\ell} P_{\ell}(x_i)}$$

- Update mixing weights, means, and covariances:

$$\pi_j = \frac{1}{n} \sum_{i=1}^n w_{ij}$$

$$\mu_j = \frac{1}{n\pi_j} \sum_{i=1}^n w_{ij} x_i$$

$$\Sigma_j = \frac{1}{n\pi_j} \sum_{i=1}^n w_{ij} (x_i - \mu_j) (x_i - \mu_j)^T$$



## Example

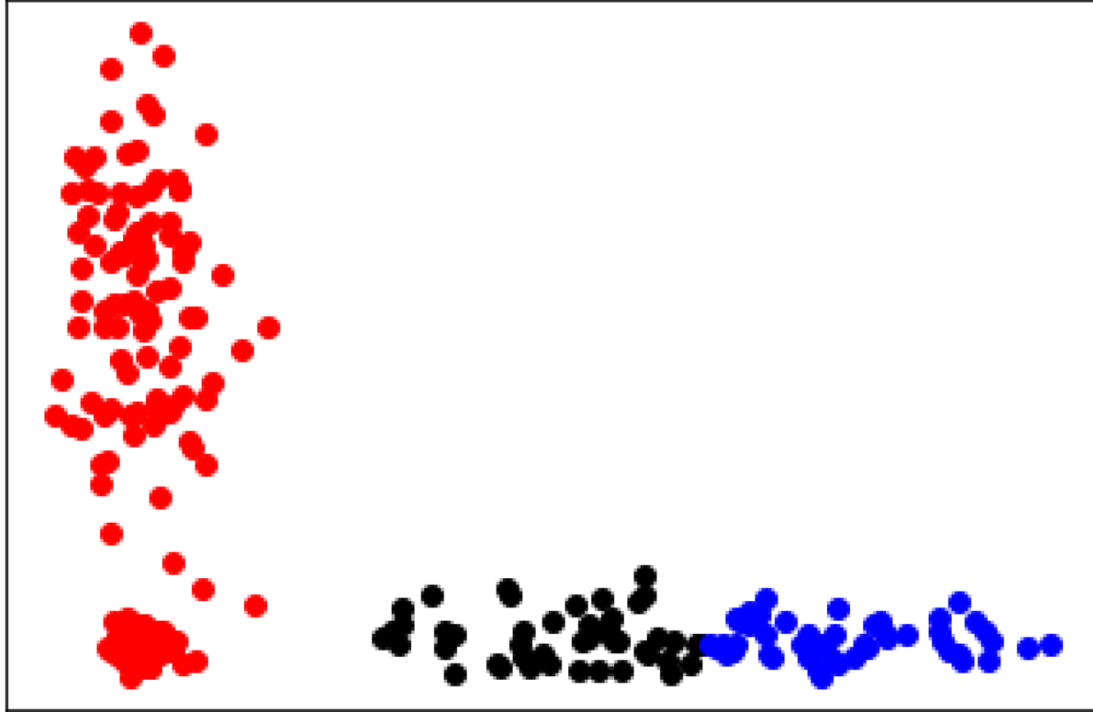
Data with 3 clusters, each with 100 points.



The data set

## Example

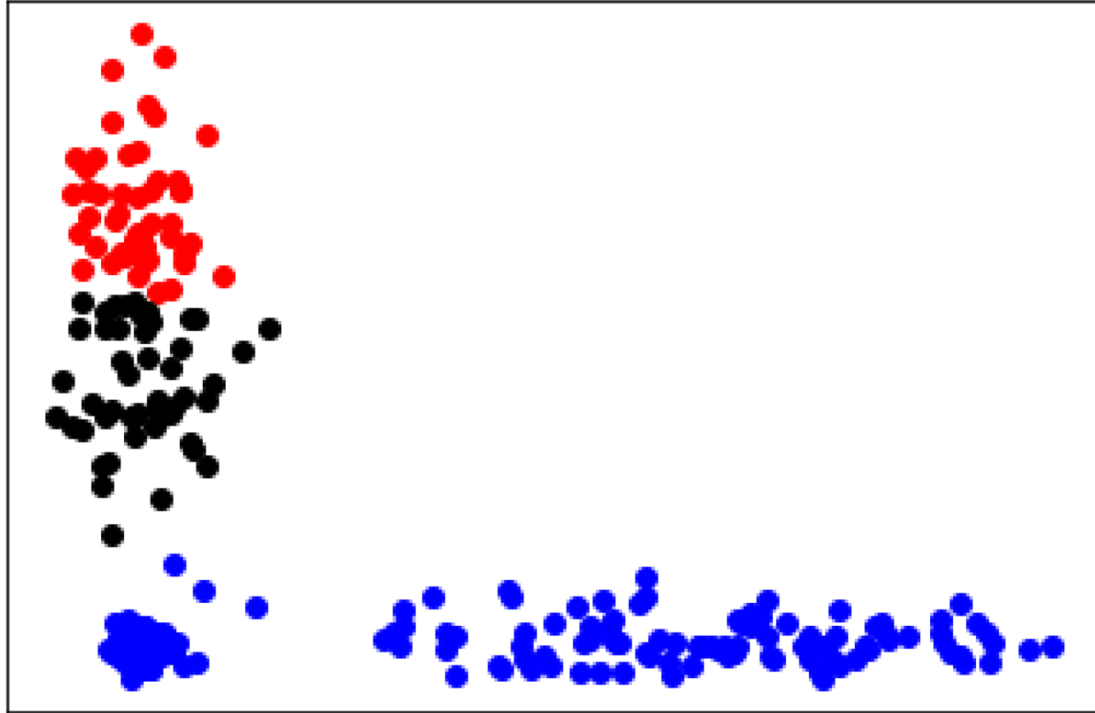
Data with 3 clusters, each with 100 points.



$k$ -means solution 1

## Example

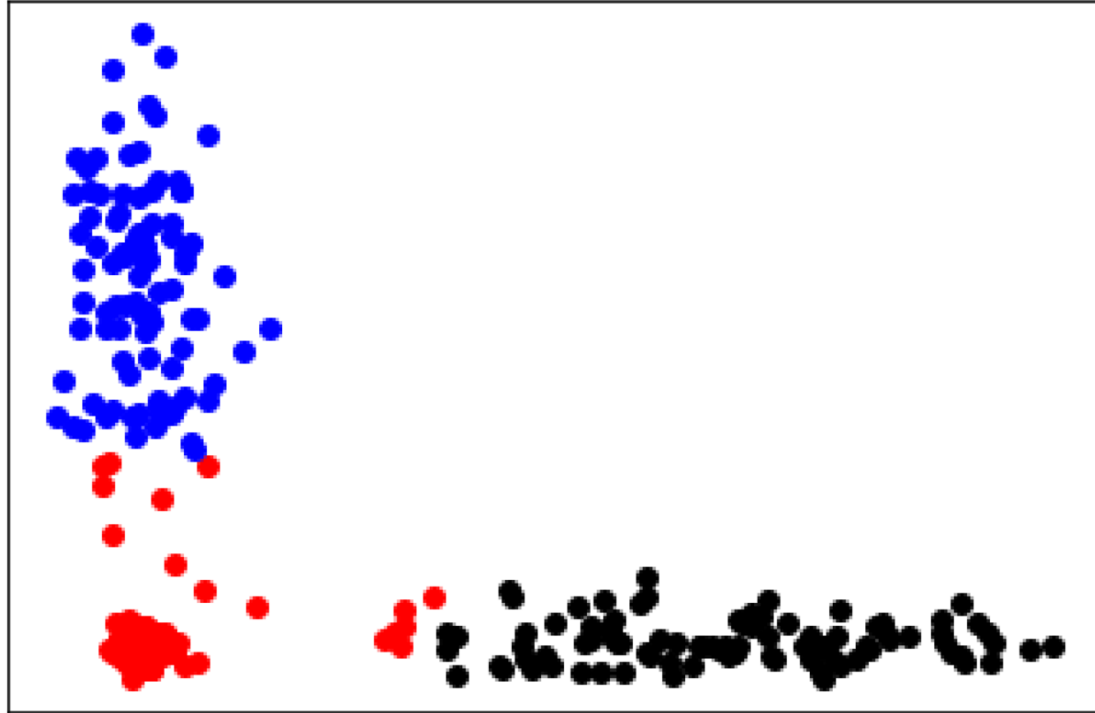
Data with 3 clusters, each with 100 points.



$k$ -means solution 2

## Example

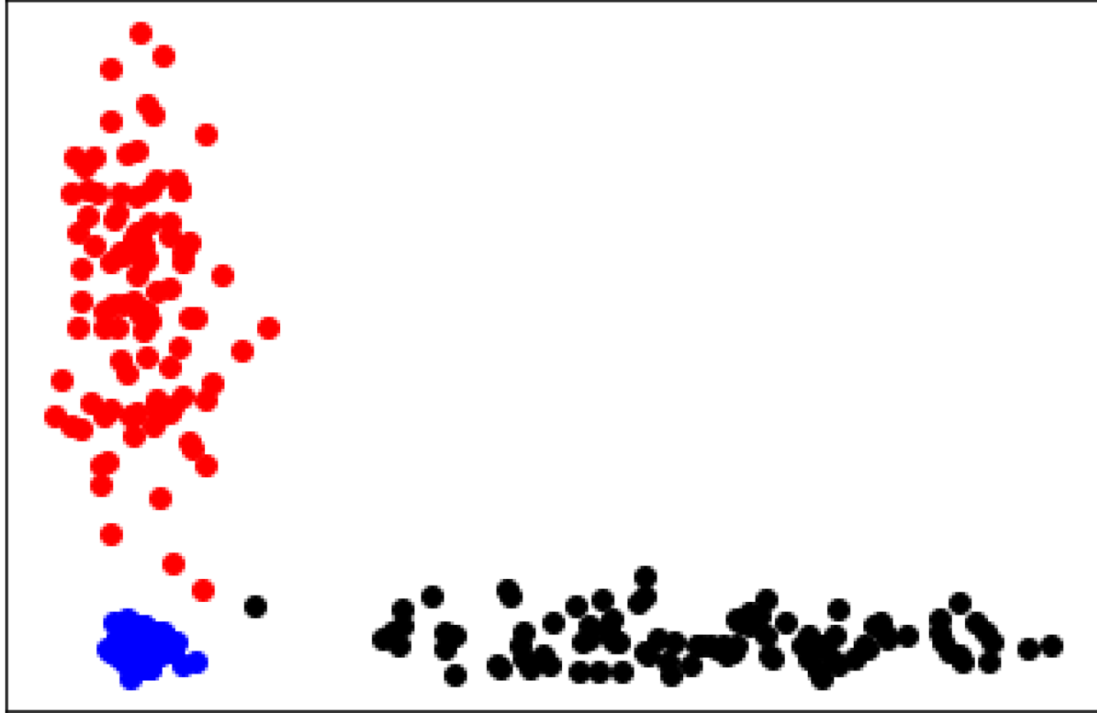
Data with 3 clusters, each with 100 points.



$k$ -means solution 3

## Example

Data with 3 clusters, each with 100 points.



EM for mixture of Gaussians